# Gromov Compactness for Genus 0 Pseudo-Holomorphic Curves

A Formalization Blueprint in Lean 4

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# Contents


1 Status and conventions 2

2 Foundations 3

3 Certificates (assumed hypotheses) 5

4 Local analysis 7

5 Concentration and bubbling 9

6 Assembly and the master theorem 11

7 Extensions 13

8 Certificate audit 14

9 Normal-chart infrastructure (N1–N3) 15

# Chapter 1

# Status and conventions

The master theorem Theorem 6.7 is a machine-checked *reduction*: it is proved in Lean with no `sorry`, from explicit hypotheses called certificates (Chapter 3). All library declarations depend only on `propext`, `Classical.choice` and `Quot.sound` (census: `tests/PhaseAxiomCensus.lean`). Certificates were audited for satisfiability in `tests/CertificateSanity.lean`; two earlier certificates were shown to be too strong and replaced (Chapter 8). Several certificates are close to, or identical to, the conclusion of the theorem they serve; this is recorded in each node.

# Chapter 2

# Foundations

**Definition 2.1** (Closed tame target)**.** A compact boundaryless smooth manifold $M$ with pointwise nondegenerate skew forms $\omega_x$, closed in charts, an almost-complex structure $J$ with $J^2 = -1$, smooth in charts, and tameness $\omega_x(v, J_x v) > 0$ for $v \neq 0$.

**Definition 2.2** (Compatible metric)**.** $g_x(v, w) = \frac{1}{2}(\omega_x(v, J_x w) + \omega_x(w, J_x v))$.

**Lemma 2.3** (Metric properties)**.** *$g$ is symmetric, positive definite and $J$-invariant.*

*Proof.* Algebra from tameness and $J^2 = -1$. □

**Definition 2.4** ($J$-holomorphic map)**.** $u$ is smooth on $s$ and $du \circ j = J \circ du$ at every point of $s$.

**Lemma 2.5** (Reparametrization invariance)**.** *$J$-holomorphicity is preserved by biholomorphic reparametrization of the domain.*

*Proof.* Chain rule for `mfderiv`. □

**Definition 2.6** (Möbius frames)**.** Möbius transformations of $\mathbb{CP}^1$, with projective compatibility (`is_moebius`), composition, inverse, and centered stereographic affine rescalings $w \mapsto a + \lambda w$ (`affine_scaling`).

**Theorem 2.7** (Three-point transitivity)**.** *Any triple of distinct points of $\mathbb{CP}^1$ can be sent to any other by a unique Möbius map.*

*Proof.* Adjugate matrices. □

**Definition 2.8** (Energy measure and total energy)**.** The push-forward from the stereographic chart of Lebesgue measure with density $\frac{1}{2}|du|_g^2 \cdot 4/(1 + |w|^2)^2$; $E(u)$ is its real total.

**Theorem 2.9** (Conformal invariance of energy)**.** *Energy is invariant under Möbius reparametrization.*

*Proof.* Change of variables with the spherical Jacobian. □

**Lemma 2.10** (A nonconstant sphere of finite positive energy)**.** $\mathrm{id} : \mathbb{CP}^1 \to \mathbb{CP}^1$ *is $J$-holomorphic with* $0 < E(\mathrm{id}) < \infty$.

*Proof.* Energy density of the identity is identically 1. □

**Definition 2.11** (Smooth local convergence)**.** $C^\infty_{\mathrm{loc}}$ convergence on an open subset of $\mathbb{CP}^1$.

**Lemma 2.12** (Closedness of $J$-holomorphicity)**.** $C^\infty_{\mathrm{loc}}$ *limits of $J$-holomorphic maps are $J$-holomorphic.*

*Proof.* Pass to the limit in $du \circ j = J \circ du$. □

**Lemma 2.13** (Diagonal extraction)**.** *A nested sequence of subsequences has a diagonal subsequence eventually inside each.*

*Proof.* Cantor diagonal argument. □

**Definition 2.14** (Stable sphere tree limit and compactness witness)**.** A finite tree of $J$-holomorphic spheres with nodal matching and stable ghosts; a witness records a subsequence, frames, $C^\infty_{\mathrm{loc}}$ convergence, frame collapse and energy convergence.

**Theorem 2.15** (Component bounds)**.** $Ph \le B$*; for* $|V| > 1$*,* $G \le P - 2$ *and* $|V| \le 2P - 2$*;* $|V| \le \max(1, 2\lfloor B/h \rfloor - 2)$.

*Proof.* Handshake lemma and the energy gap packaged in `TargetThresholds`. □

# Chapter 3

# Certificates (assumed hypotheses)

**Proposition 3.1** (Certificate `h_fin`: finite energy)**.** *Each $u_n$ has finite extended energy. (Needed because $E$ is a `toReal`; to be discharged in milestone N1.)*

*Proof.* Not formalized (planned: Theorem 9.2). □

**Definition 3.2** (Universal thresholds)**.** A bundle of constants $\varepsilon_{\mathrm{reg}}, C_{\mathrm{reg}}, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}, C_{\mathrm{neck}}, a, h$ together with the estimates they satisfy ($\varepsilon$-regularity, energy gap, neck decay).

**Proposition 3.3** (Certificate: existence of thresholds)**.** *For the given target, `TargetThresholds` is inhabited. The master theorem takes a threshold bundle as a parameter. Theorem 4.13 constructs one from Propositions 3.4, 3.5 and 3.7.*

*Proof.* Not formalized unconditionally. □

**Proposition 3.4** (Certificate `h_weitz`: Weitzenböck decomposition)**.** *In stereographic coordinates the energy density $e$ of a $J$-holomorphic sphere is smooth and $\Delta e = |\cdot|^2 + K$ with $|K| \le be^2$. Essentially equivalent to the conclusion of Lemma 4.2.*

*Proof.* Not formalized (planned: N3, normal charts). □

**Proposition 3.5** (Certificate `h_hit` / `h_hit_exit`: isoperimetric bound and ball exit)**.** $r^2 \le E(u; u^{-1}B_r(x))$ *at boundary-hit radii, and nonconstant spheres exit small balls. Close to the conclusion of Theorem 4.7.*

*Proof.* Not formalized (planned: N3). □

**Proposition 3.6** (Certificate `h_osc`: collar oscillation)**.** *On the collar $A(2r, R/2)$ the oscillation of $u$ is at most $4\sqrt{E(u; A(r, R))}$.*

*Proof.* Not formalized (planned: N3). □

**Proposition 3.7** (Certificate `h_decay`: exponential decay on cylinders)**.** $E(u; A(r_1, r_2)) \le C_0((r/r_1)^a + (r_2/R)^a)E(u; A(r, R))$ *below $\varepsilon_{\mathrm{neck}}$. Identical to the conclusion of Theorem 4.10.*

*Proof.* Not formalized (planned: N3). □

**Proposition 3.8** (Certificate `h_ext`: extension across a puncture)**.** *A finite-energy punctured $J$-holomorphic sphere has a continuous weak $J$-holomorphic extension. Identical to the conclusion of Theorem 4.11.*

*Proof.* Not formalized (planned: N3). □

**Proposition 3.9** (Certificate `h_tower`: regularity tower)**.** *Weak $J$-holomorphic maps are $C^n$ for every $n$. Implies the conclusion of Theorem 4.5 trivially.*

*Proof.* Not formalized (planned: N3, Schauder in normal charts). □

**Proposition 3.10** (Certificate `h_aa`: Arzelà–Ascoli extraction)**.** *Under local gradient bounds, a subsequence converges in $C^\infty_{\mathrm{loc}}$. Identical to the existence part of Theorem 4.6.*

*Proof.* Not formalized (planned: N3). □

**Proposition 3.11** (Certificate `h_conc_extract`: smooth extraction off the singular set)**.** *Along a further subsequence the maps converge in $C^\infty_{\mathrm{loc}}$ off $S$ to a $J$-holomorphic $u_0$, and $\mu = \mu_{u_0} + \sum_{p\in S} \mu\{p\}\delta_p$.*

*Proof.* Not formalized. □

**Definition 3.12** (Genuine defect of a sequence)**.** $f$ is $J$-holomorphic with bounded finite energy, $\lim_{r\to 0} \lim_n E(f_n; D_r(a)) = m$, and near $a$ the maps eventually stay close to $b$.

**Proposition 3.13** (Certificate `OneLevelCertificate` (Contract B04, sequence-relative))**.** *Every genuine defect of mass $m \geq h$ admits a realized one-level decomposition (subsequence, moving centers and scales, Möbius rescaling frames, frame collapse, $C^\infty_{\mathrm{loc}}$ convergence off nodes, and genuine child defects).*

*Proof.* Not formalized (this is the analytic core of bubbling). □

**Proposition 3.14** (Certificate `ConcDefectCertificate`)**.** *Every atom of the concentration measure is a genuine defect of a rotated subsequence of $u_{\sigma(n)}$, with its true mass and base value $u_0(p)$.*

*Proof.* Not formalized. □

**Proposition 3.15** (Extension certificates)**.** *Three-point rigidity of homeomorphisms (`h_rigidity`), finiteness of component symmetries (`h_finite_symm`), metrizability of the moduli space (`h_metr`), and the limit extraction certificates for marked curves, moduli space and varying $J$ (`h_extract`).*

*Proof.* Not formalized. □

# Chapter 4

# Local analysis

**Lemma 4.1** (Scalar mean-value inequality (A01)). *If $e \geq 0$ is $C^2$ with $\Delta e \geq -be^2$ on $D_r$ and $\int_{D_r} e < \eta$, then $\sup_{D_{r/2}} e \leq Cr^{-2} \int_{D_r} e$.*

*Proof.* Heinz-type argument in flat coordinates. □

**Lemma 4.2** (Bochner inequality (A02)). *$\Delta e \geq -be^2 - ce$ for the energy density of a $J$-holomorphic sphere.*

*Proof.* Algebra from Proposition 3.4. □

**Theorem 4.3** ($\varepsilon$-regularity (A03)). *Small energy on a disk bounds the energy density on the half disk.*

*Proof.* Mean-value inequality applied to the Bochner inequality. □

**Lemma 4.4** (Flat elliptic estimates (A04)). *Interior Cauchy–Riemann estimates, 2D Morrey–Sobolev embedding, Cauchy–Pompeiu Hölder gain and one Hölder bootstrap step, in flat coordinates.*

*Proof.* Mathlib's Cauchy integral formula and a near/far splitting. □

**Theorem 4.5** (Nonlinear bootstrap (A05)). *Weak $J$-holomorphic maps are smooth.*

*Proof.* Immediate from Proposition 3.9. □

**Theorem 4.6** (Local smooth compactness (A06)). *Under local gradient bounds a subsequence converges in $C^\infty_{\mathrm{loc}}$ to a $J$-holomorphic limit.*

*Proof.* Extraction from Proposition 3.10; limit is $J$-holomorphic by Lemma 2.12. □

**Theorem 4.7** (Target monotonicity (A08)). *$c r^2 \leq E(u; u^{-1}B_r(x))$ for small $r$ when $u$ leaves $B_r(x)$.*

*Proof.* From Proposition 3.5. □

**Theorem 4.8** (Sphere energy gap (A07)). *Nonconstant $J$-holomorphic spheres have energy at least $\varepsilon_{\mathrm{sph}} > 0$.*

*Proof.* Monotonicity at a ball the sphere must exit. □

**Theorem 4.9** (Cylinder localization (A09)). *Small annular energy confines the collar image to a small ball.*

*Proof.* From Proposition 3.6. □

**Theorem 4.10** (Exponential decay on cylinders (A10)). *Annular energy decays exponentially away from the collars.*

*Proof.* Restates Proposition 3.7. □

**Theorem 4.11** (Extension across a puncture (A11)). *A finite-energy punctured $J$-holomorphic sphere extends continuously and weakly $J$-holomorphically.*

*Proof.* Restates Proposition 3.8. □

**Theorem 4.12** (Removal of singularities (A12)). *The extension is a unique smooth $J$-holomorphic sphere.*

*Proof.* Bootstrap of the extension; uniqueness by density of the punctured sphere. □

**Theorem 4.13** (Thresholds from analytic certificates (B01)). *Propositions 3.4, 3.5 and 3.7 yield a threshold bundle with $0 < h \leq \min(\varepsilon_{\mathrm{reg}}/2, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}/2)$.*

*Proof.* Take minima of the constants. □

# Chapter 5

# Concentration and bubbling

**Theorem 5.1** (Prokhorov compactness on $\mathbb{CP}^1$)**.** *Bounded sequences of finite measures on $\mathbb{CP}^1$ have weakly convergent subsequences.*

*Proof.* Diagonal argument over a dense sequence in $C(\mathbb{CP}^1)$ and Mathlib's `RealRMK.rieszMeasure`. □

**Theorem 5.2** (Weak compactness of energy measures (C01))**.** *Energy measures of a bounded-energy sequence have a weakly convergent subsequence.*

*Proof.* From Theorem 5.1. □

**Definition 5.3** (Concentration data)**.** A subsequence, a limit measure $\mu$, a finite singular set $S$ and a $J$-holomorphic limit $u_0$ off $S$.

**Theorem 5.4** (Existence of concentration data (C02))**.** *Every bounded-energy sequence admits concentration data.*

*Proof.* Weak compactness, finiteness of $S$, and Proposition 3.11. □

**Theorem 5.5** (Defect identity (C03))**.** $\mu = \mu_{u_0} + \sum_{p \in S} m_p \delta_p$ *with* $m_p \geq \varepsilon_{\mathrm{sph}}$*, and* $E(u_{\sigma(n)}) \to E(u_0) + \sum m_p$*.*

*Proof.* Finite-measure Portmanteau. □

**Lemma 5.6** (Calibrated rescaling (B02))**.** *Hofer's lemma specialized to the plane gives centers and scales with $E(f_n; D_{\lambda_n}(a_n)) = m - h/2$.*

*Proof.* From `Mathlib.Analysis.Hofer`. □

**Theorem 5.7** (Neck vanishing (B03))**.** *Neck energy and neck diameter tend to zero with the stated quantifier order.*

*Proof.* From cylinder localization and Proposition 3.6. □

**Definition 5.8** (One-level decomposition)**.** A bubble $v$, child points $q$ and masses with $m = E(v) + \sum m_i$, $v(\infty) = b$, and calibration.

**Lemma 5.9** (Ghost branching (B05))**.** *A constant bubble has at least two children.*

*Proof.* Calibration arithmetic. □

**Lemma 5.10** (Strict rank decrease (B06))**.** $\lfloor m_i/h \rfloor < \lfloor m/h \rfloor$*; well-founded induction on this rank.*

*Proof.* Children have mass at most $m - h$. □

**Definition 5.11** (Local and realized defect trees)**.** A finite stable tree of $J$-holomorphic spheres of total energy $m$ attached at $b$; a realized local defect tree also records the subsequence, per-vertex Möbius frames, transition collapse and $C^\infty_{\mathrm{loc}}$ convergence off incident nodes.

**Theorem 5.12** (Resolution of a genuine defect (B07))**.** *Every genuine defect of mass $m > 0$ is resolved by a realized local defect tree.*

*Proof.* Energy-rank induction; realized children are grafted sequentially along composed subsequences and telescoped frames (`graft_child_defect_trees_realized`). □

# Chapter 6

# Assembly and the master theorem

**Lemma 6.1** (Global frames (G01))**.** *Rescaling data on a rooted tree determine global Möbius frames with the transition relations.*

*Proof.* Well-founded recursion on depth. □

**Definition 6.2** (Grafted tree and prestable limit data)**.** A tree rooted at $u_0$ with defect trees attached at $S$; realized grafted trees and prestable limit data carry the subsequence, Möbius frames, transition collapse and $C^\infty_{\mathrm{loc}}$ convergence.

**Theorem 6.3** (Realized root-sphere grafting theorem)**.** *Grafting realized local defect trees onto the root curve $u_0$ across all concentration atoms $p \in S$ produces a realized grafted tree with global frame collapse and $C^\infty_{\mathrm{loc}}$ convergence off nodes.*

*Proof.* Finite-set induction over $s \subseteq S$, composing subsequences and rotating/telescoping child frames. □

**Theorem 6.4** (Prestable limit (G01))**.** *Every bounded-energy sequence of $J$-holomorphic spheres has prestable limit data.*

*Proof.* Concentration data, defect identity, realized B07 resolution at each atom of $S$ via Propositions 3.13 and 3.14, and Theorem 6.3. □

**Theorem 6.5** (Stabilization (G02))**.** *Unstable ghost roots can be pruned or suppressed, yielding a stable limit.*

*Proof.* Strong induction on $|V|$. □

**Theorem 6.6** (Core compactness (G03))**.** *A subsequence converges to a stable sphere tree limit.*

*Proof.* Assembly followed by stabilization. □

**Theorem 6.7** (Genus-zero Gromov compactness)**.** *Main theorem of the project, as a reduction to the certificates of Chapter 3.*

*Proof.* Core compactness plus component bounds. □

# Chapter 7

# Extensions

**Theorem 7.1** (Varying almost-complex structures (E01))**.** *Limits of $J_n$-holomorphic maps with $J_n \to J$ are $J$-holomorphic (unconditional); compactness for varying $J$ (from Proposition 3.15).*

*Proof.* Operator-norm squeeze; extraction certificate. □

**Theorem 7.2** (Marked compactness (E02))**.** *Compactness for marked stable maps.*

*Proof.* From the marked extraction certificate. □

**Theorem 7.3** (Automorphisms (E04))**.** *Möbius three-point rigidity (unconditional); stability iff finite automorphisms (from Proposition 3.15).*

*Proof.* Fibrewise finiteness. □

**Theorem 7.4** (Moduli space (E05))**.** *$\overline{\mathcal{M}}_{0,k}(M, J, \leq B)$ and its class-$A$ part are compact Hausdorff.*

*Proof.* Sequential compactness plus metrizability certificate. □

# Chapter 8

# Certificate audit

*Remark* 8.1 (Withdrawn certificates and definition repairs). Up to commit `0a3de26` the master theorem assumed `h_one_level` (a decomposition for *every* mass $m \geq h$) and `h_graft_extract` (convergence to *every* grafted tree with arbitrary masses). `tests/CertificateSanity.lean` proves: S2, the former forces a sphere of energy exactly $m$ for each $m \in [h, 2h)$ and so fails on any target missing one such value (e.g. $\mathbb{CP}^1$); S4, the latter is contradictory for sequences with convergent energies and two admissible trees. S1 checks that $u_n = \mathrm{id}_{\mathbb{CP}^1}$ satisfies the non-certificate hypotheses, S3 that Definition 3.12 fails without energy concentration, S5 that `affine_scaling` genuinely translates by its center $a \in \mathbb{C}$, and S6 that `SmoothLocalConvergesOn` holds for the nonconstant identity map on $\mathbb{CP}^1$. Both over-strong certificates were replaced by Propositions 3.13 and 3.14, and `GraftCertificate` was eliminated in favour of the proved theorem Theorem 6.3.

# Chapter 9

# Normal-chart infrastructure (N1–N3)

**Theorem 9.1** (Finite good atlas and `mfderiv` factorization (N1))**.** *A compact manifold has a finite atlas with a positive Lebesgue radius, and* ***`mfderiv`*** *factors through its charts via* ***`tangentCoordChange`****.*

*Proof.* Finite subcover of precompact coordinate balls and Lebesgue number lemma; chain rule for `extChartAt` and `tangentCoordChange_comp`. □

**Theorem 9.2** (Energy density certificate and `h_fin` discharge (N1))**.** *Bounded Riemannian energy density on $\mathbb{CP}^1$ implies finite extended energy (Proposition 3.1).*

*Proof.* Comparison with the finite area of $\mathbb{CP}^1$ under the spherical Jacobian. □

**Theorem 9.3** (Quadratic Taylor normal charts, Koszul cancellation and $\mathbb{CP}^1$ model (N2))**.** *Quadratic Taylor normalization $\Phi_{y_0}(v) = y_0 + v - \frac{1}{2}\Gamma_{y_0}(v, v)$ yields local diffeomorphisms via the inverse function theorem; composing with atlas charts produces centered charts $\Psi_p$ with $\Psi_p(0) = p$, $d\Psi_p(0) = \mathrm{id}$, and algebraic Koszul cancellation of the first metric derivatives at 0.* ***`UniformNormalChartFamily`*** *packages the 15 geometric fields (uniform radius, bi-Lipschitz distance comparison, curvature and $C^3$ tensor bounds) and is unconditionally inhabited on $(\mathbb{CP}^1, \omega_{\mathrm{FS}}, J_{\mathbb{CP}^1})$ via* ***`concrete_CP1`****.*

*Proof.* `ContDiffAt.toOpenPartialHomeomorph`, Koszul symmetry cancellation, and explicit stereographic bi-Lipschitz bounds on $\mathbb{CP}^1$. □

**Theorem 9.4** (Local-chart certificate bridge and coordinate PDE building blocks (N3))**.** *Given a* ***`UniformNormalChartFamily`*** *and conditioned* ***`UniformNormalChartCertificates`*** *(where oscillation, decay and puncture removal are conditioned on $J$-holomorphicity and non-degenerate radii, and proved inhabited on constant curves via* ***`conditioned_certificates_inhabited_of_all_const`****), the bridge theorems instantiate Propositions 3.4 to 3.10. The coordinate-level building blocks (centered normal chart Hessian expansion and quadratic $K$-tensor bound, polar slice FTC and quadratic energy growth from the radial differential inequality, Widman hole-filling geometric decay and collar diameter bound, Cauchy filter extension across a puncture, quasilinear Laplacian commutator bound $|\Delta v| \le C|Dv|^2$, and dense-set Arzelà–Ascoli extraction) are also proved in* ***`Analysis/`****.*

*Proof.* Projections, degenerate-collar splitting, curvature constant combination, and coordinate calculus. □